

Final Examination on Calculus 2 - 2022.2

Elitech programs

Total time: 90 minutes

Attention: No document is allowed.

Prob 1. (1 point) Find the curvature of the curve $y = x^2 - 1$ at $A(1, 0)$.

Prob 2. (1 point) Find the directional derivative of the function $u(x, y, z) = x^2 + 2xy^2 - yz^3$ in the direction of $\vec{l} = (1, -2, 2)$ at the point $A(1, 1, 1)$.

Prob 3. (1 point) Evaluate $\iint_D (1 + x + y^2) dx dy$, where $D : x^2 + y^2 \leq 1$.

Prob 4. (1 point) Evaluate $\iiint_V (3x + z) dx dy dz$, where $V : x^2 + y^2 + z^2 \leq 2z$.

Prob 5. (1 point) Evaluate $\int_0^{+\infty} \frac{\sqrt{x}}{(1+x^2)^3} dx$.

Prob 6. (1 point) Evaluate $\int_C x + 2y ds$, where $C: y = \sqrt{2x - x^2}$.

Prob 7. (1 point) Evaluate $\oint_C (xy + 3x + 2y) dx + (xy - 2y) dy$, where C is the circle $x^2 + y^2 = 2x$ with counterclockwise orientation.

Prob 8. (1 point) Evaluate $\iint_S (2x - y + z^2) dS$, where S is the hemisphere $S : x^2 + y^2 + z^2 = 1, x \geq 0$.

Prob 9. (1 point) Find the flux of the vector field $\vec{F} = x\vec{i} + y\vec{j} + (z^2 - 1)\vec{k}$ across S , where S is the part of the ellipsoid $x^2 + \frac{y^2}{4} + z^2 = 1, z \geq 0$, with upward orientation.

Prob 10. (1 point) Find the circulation of the vector field

$$\vec{F} = (2xze^{x^2} + y^2 - z)\vec{i} + (y - 3z)\vec{j} + (e^{x^2} + x + 2y)\vec{k}$$

around C . Here C is the curve of intersection of the plane $x + y + z = 1$ and the cylinder $x^2 + y^2 = 2y$, oriented counterclockwise as viewed from above.

Chúc các bạn hoàn thành tốt bài thi